

Query Taxonomy Validation — Findings

1. Objective

Validate whether the current manual taxonomy reflects how users actually think about college-related queries.

Dataset — 6 sources

- College features
- Student questions
- Parent questions
- HS counselor queries
- Academic advisor queries
- B2B platform queries + consumer search dumps

Method

- TF-IDF vectorization → K-means (k=20, selected by silhouette score)
- Dimensionality reduction via SVD/LSA
- LLM-assisted cluster labeling → manual review

2. Clusters Summary

Cluster	Size	Notes / Issues
Student Life & Academics (general)	473	Very mixed; needs splitting
College-Life Questions (catch-all)	149	Making friends, housing, FOMO, dating, first-gen navigation
College Features: Programs & Research	115	Attribute-based, e.g., research labs, majors
College Features: School-Level Attributes	111	Culture, policies, campus features
College Features: University-Level	71	Same as above; should merge with school-level
B2B Platform Partner Queries	68	CRM, API, pricing, demo; distinct user type
HS Counselor + Advisor: Student Support	64	Disabilities, anxiety, first-gen support
HS Counselor: Application Advising	64	Essays, waitlists, list balancing
Engineering & CS Programs	52	Co-ops, weed-out classes, internships; noisy queries
Career Outcomes & Rankings	43	Best colleges for careers, feeder schools
Admissions Fit & General Advice	40	GPA cutoffs, prestige vs fit, test-optional
Financial Aid & Cost	34	FAFSA, CSS, merit aid, loans
College Features: Specific Academic Strengths	34	Overlaps Programs & Research; should merge
Advisor: Individual Student Decision-Making	34	Debt, withdrawals, transfers, mental health
Applicant Vibe & Social Fit	34	Campus social scene, fun vs prestige
First-Year Experience & Belonging	32	Freshman challenges, Greek life, orientation

Cluster	Size	Notes / Issues
Student-Run Clubs & Activities	31	Overlaps Student-Led Initiatives
Student-Led Initiatives	25	Near-duplicate; merge with clubs
Mental Health & Wellness	18	Counseling, burnout, therapy costs
Active Clubs & Orgs	15	Merge with student-run/student-led clusters

3. Key Insights

1 — Fragmentation in College Features Queries

Many queries are semantically identical but split by surface wording. E.g., "student-run radio stations" vs "active student radio" — these should collapse into Campus Life & Activities.

2 — B2B Queries Are Distinct

Platform, integrations, pricing — entirely separate from student queries. Needs its own category or pipeline.

3 — "College Search" Varies by Persona

Intent differs across students, parents, and admissions staff. Persona labeling would improve result relevance.

4 — Advisor & Counselor Queries Are Too Broad

The current single cluster conflates very different needs. Recommend splitting into:

- Individual student support
- Degree planning
- Institutional / policy advising

5 — Engineering Dataset Is Noisy

Abbreviated or malformed queries are hurting cluster quality. Needs a cleaning pass before re-analysis.

4. Recommended Taxonomy Changes

A — Merge Similar Categories

- Student-run + student-led + active clubs → Campus Life & Activities
- College programs + school-level features + university-level features → College Features & Attributes
- Specific academic strengths → fold into College Features & Attributes

B — Split Overly Broad Categories

Student Life & Academics →

- Academic Navigation
- Social & Campus Life
- Finances & Logistics
- Post-Grad Planning

Academic Advisor →

- Individual Student Support
- Degree Planning & Requirements
- Institutional & Policy Advising

HS Counselor →

- Application Advising
- Student Support & Special Populations

C — Add a New Top-Level Category

B2B / Institutional Partners — covering Platform Features, Data & Integrations, Pricing & Contracts, Recruitment Strategy.

D — Introduce Persona Tags

Apply across all categories without restructuring core taxonomy: Student, Parent, Counselor, Advisor, Partner.

5. Proposed Revised Taxonomy

Category	Sub-Categories
College Discovery & Search	Features & Attributes, Academic Programs, Campus Culture, Location & Setting, Vibe & Fit
Admissions Process	Application Strategy, Essay Help, Test Prep, Timelines, Financial Aid Navigation
Student Academic Life	Major Selection, Degree Planning, Course Navigation, GPA & Grades, Research & Internships
Student Social & Campus Life	Making Friends, Housing, Greek Life, Clubs & Activities, Wellness
Student Finances	FAFSA, Scholarships, Loans, Budgeting, Work Study
Post-Grad Planning	Graduate School, Professional School, Career Paths, First Jobs
Parent Perspective	Financial Planning, Campus Safety, Supporting Your Student, Admissions as a Parent
HS Counselor Practice	Application Advising, Student Support, Special Populations, Counselor Tools
College Academic Advising	Individual Student Support, Degree Planning, At-Risk Students, Institutional Policy
B2B / Institutional Partners	Platform Features, Data & Integration, Pricing & Contracts, Recruitment Strategy

6. Next Steps

1. **Clean engineering dataset** — remove/fix abbreviated and malformed queries
2. **Add persona layer** — identify who is asking each query
3. **Validate updated categories** — manually label ~200 sample queries
4. **Improve query grouping** — use semantic clustering to consolidate similar queries
5. **Finalize category structure** — decide if Mental Health & Wellness stays separate or merges under Student Social & Campus Life